# MS MARCO Generative QA — Progress Report

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## Project goal

Build an end-to-end open-domain QA system on MS MARCO that goes **query → retrieve passages → generate an answer**, and measure each stage in isolation so it's clear where errors come from. The aim is not SOTA; the aim is a clean, reproducible pipeline whose retrieval, reranking, and generation stages can each be ablated.

## Methods (by week)

| Week | Stage | Data | Method |
|---|---|---|---|
| W1 | EDA | MS MARCO QA v2.1 validation, 5k-row sample | Length / type / answer-type distributions; sizing decisions for W2–W5 |
| W2 | Sparse retrieval (BM25) | MS MARCO Passage (8.8M docs), dev/small (6 980 q) | `bm25s` backend, `k1=1.5`, `b=0.75`, top-1000 per query |
| W3 | RAG generation | BM25 / reranked top-3 ∩ QA references, full dev/small (6 980 q) | T5-small (no fine-tuning), prompt = `question: <q> context: <p1 p2 p3>` |
| W4 | Dense retrieval | Qrels-anchored 50k sample of Passage, dev/small (6 980 q) | `all-MiniLM-L6-v2` + FAISS `IndexFlatIP`; same sample re-indexed with BM25 for apples-to-apples comparison |
| W5 | Cross-encoder reranking | W4 dense top-100, full dev/small (6 980 q) | `cross-encoder/ms-marco-MiniLM-L-6-v2` rerank, depth K=100 |

All runs share a single config (`configs/baseline.yaml`); the four entrypoints in `experiments/run_*.py` produce the metrics each report cites.

# Key results

**W1 (EDA)** — Queries are short (5–10 words median); passages ~10× longer (median ~50 words). `DESCRIPTION` and `NUMERIC` dominate the query-type buckets; a non-trivial fraction of queries has no answer and must be filtered for any SFT supervision. Most queries have 0 or 1 marked-relevant passage (sparse, binary qrels).

**W2 (BM25 baseline, full 8.8M corpus, 6 980 dev/small queries)**

| Metric | Value |
|---|---|
| MRR@10 | 0.1703 |
| Recall@100 | 0.6212 |
| Recall@1000 | 0.8154 |

Published Anserini/Lucene reference is ~0.184 MRR@10; the ~0.014 gap is attributable to tokenizer differences between Lucene and `bm25s`. No `k1`/`b` tuning. Search cost: 586 ms/query.

**W3 (RAG, T5-small + retrieved top-3, full dev/small 6 980 queries)**

Two paired runs over the **same 6 980 dev/small qids** (mutually restricted via `--restrict-to-run`), differing only in upstream retrieval source:

| Retrieval → T5-small | ROUGE-L | BLEU | EM | Token-F1 |
|---|---:|---:|---:|---:|
| BM25 top-3 | 0.1859 | 0.0717 | 0.0135 | 0.1966 |
| Reranked top-3 | 0.3621 | 0.2922 | 0.0606 | 0.3677 |
| **Δ (rerank − BM25)** | **+0.1763** | **+0.2206** | **+0.0471** | **+0.1711** |

Reranking the first stage **roughly doubles every generation metric** on the full benchmark. 95 % paired-bootstrap CIs (n = 6 980, 10 k resamples, seed 42) on the per-query Δ all sit strictly above zero: ROUGE-L [+0.1663, +0.1820], BLEU [+0.1265, +0.1395], EM [+0.0417, +0.0527], Token-F1 [+0.1632, +0.1789]; two-sided $p < 0.001$ in every case.

Mechanism (retrieval flag): the rate of having at least one relevance-judged passage in the top-3 jumps from **20.8 % (BM25) to 96.9 % (reranked)** — the cross-encoder is what gets the relevant passage into the generator's window.

By query type (token-F1 Δ), `DESCRIPTION` (n=3725) gains most (+0.2050); `NUMERIC` (n=1665) gains least (+0.1238) — short numeric answers depend more on lexical surface match than on which-of-the- near-duplicates the reranker picks. Per-query split: **4 015 improvements / 1 766 regressions / 1 199 ties** over 6 980 queries.

T5-small is pretrained-only (no SFT) so absolute level numbers are modest. The load-bearing claim is the **paired Δ between rows**, which is statistically robust on n = 6 980.

**W4 (Dense vs BM25 on the same 50k qrels-anchored sample, 6 980 queries)**

| Metric | BM25 (sample) | Dense (sample) | Δ |
|---|---|---|---|
| MRR@10 | 0.6948 | 0.8830 | **+0.1882** |
| nDCG@10 | 0.7270 | 0.9041 | +0.1771 |
| Recall@100 | 0.9338 | 0.9946 | +0.0608 |
| Recall@1000 | 0.9686 | 0.9991 | +0.0305 |

Dense wins on every metric. Caveat: every dev/small relevant doc is unconditionally included in the 50k pool, so absolute numbers are upper-bounded; the *delta* between dense and BM25 on the same pool is what's meaningful. Recall@100 = 0.9946 means dense has effectively hit the ceiling on this sampled setting → from W5 onward, recall is no longer the bottleneck.

**W5 (Cross-encoder rerank over W4 dense top-100, full 6 980-query dev/small)**

| Metric | Dense (top-100) | Dense + CE rerank | Δ |
|---|---|---|---|
| MRR@10 | 0.8830 | 0.9304 | **+0.0474** |
| nDCG@10 | 0.9041 | 0.9434 | +0.0393 |
| Recall@100 | 0.9946 | 0.9946 | +0.0000 |

Recall@100 is unchanged by construction (the reranker only re-orders the top-100). The interesting columns are MRR@10 and nDCG@10: even with recall saturated, the cross-encoder still buys roughly +0.04 by tightening *local ordering* — pushing the relevant passage from rank 2–3 to rank 1. Cost: 4 h 37 min wall-clock for 538 k pairs (~32 pairs/s), peak RSS 3.3 GiB; the CE is the slowest stage by ~2 orders of magnitude. The earlier 1 000-query subsample of dev/small produced consistent deltas (MRR Δ +0.0435, nDCG Δ +0.0398), confirming the gain is not an artefact of the subsample.

# Limitations

- **Sampled corpus from W4 onward.** Dense + rerank numbers are on a 50k qrels-anchored sample where every relevant doc is present by construction. Comparable *within* W4/W5; the W4/W5 absolute MRR is *not* directly comparable to the W2 full-corpus number.
- **No fine-tuning anywhere.** Generic encoders/generator used as-is (`all-MiniLM-L6-v2`, `ms-marco-MiniLM-L-6-v2`, `t5-small`). A MS-MARCO-tuned encoder (`msmarco-MiniLM-L6-cos-v5`) or SFT generator would close most of the gap to leaderboard baselines.
- **Dev/small only, single seed.** No held-out test split; distractor sample is deterministic but not averaged across seeds.
- **No fine-tuning of T5-small.** W3 absolute level numbers (ROUGE-L ~0.18 BM25 → ~0.36 reranked) are modest because the generator is pretrained-only; the Δ between paired rows is the load-bearing signal. A SFT pass on `(question, gold passage, answer)` triples is in scope for a future week.
- **Surface-form generation metrics only.** ROUGE/BLEU/EM penalise valid paraphrases — a semantic metric (BERTScore) would give a fairer signal.

## Next steps

- Rerank BM25 top-100 (W2 run) and compare the delta to the dense delta — does the reranker recover more from a weaker first stage?
- Hybrid first-stage (RRF over BM25 + dense) → rerank.
- Scale the dense sample from 50k to 200k–500k passages to track how the BM25?dense gap shifts as relevant-doc density drops.
- Add a held-out test split and BERTScore for generation, so improvements aren't overfit to dev/small or to surface-form metrics.

## How this maps to the repo

- Conclusions per week: `reports/generated/weekNN_*.pdf`
- Process logs per week: `notebooks/weekNN_*.ipynb`
- Reproducible pipeline: `experiments/run_*.py` + `src/` + `configs/baseline.yaml`
- Reproduce locally: `make` targets (see `Makefile`); outputs land in `outputs/` (not committed).